Budget Front Desk

An agent built from public business pages, measured before and after

2026-11-10

The deliverable

A live AI front desk for Budget Heating, Cooling & Plumbing, built from a **25-field ledger** with page-and-line provenance. It triages emergency, diagnostic and estimate requests, uses the published catalogue, collects dispatcher intake, and routes to the booking page or phone. The repository contains the ledger, transcripts, graders, failures and reproducible scripts.

Before and after

	A0 baseline	A5 shipped
single-turn pass@1	113/159 (71.1%)	145/165 (87.9%)
strict pass³, shared 53 scenarios	26/53	**43/53**
integrity-gate scenarios	0/53	0/55
multi-turn, 54 conversations	36/54 (67%)	**41/54 (76%)**
multi-turn scripts passing all trials	9/18	**12/18**

The tuned arm is better on the scenarios it was iterated against: 71.1% to 87.9%, with the paired strict-pass comparison b=18, c=1, p<0.001. The multi-turn baseline now exists: A0 is 36/54 and A5 is 41/54. A5 gains 7 conversations and loses 2; McNemar p=0.180. The measured direction is positive, but the multi-turn delta is not statistically settled. These are three trials of 18 scripts, not 54 independent customers; one gain is A0's unfilled-placeholder integrity failure, so the clean behavioural signal is narrower than 7 versus 2.

What changed

The largest multi-turn gains are **arithmetic (2/9 to 5/9)** and **pushback (5/9 to 7/9)**. Memory stays 9/9 in both arms; recovery stays 9/9; change-of-mind stays 3/9. One A0 failure emitted an unfilled template placeholder; A5 passed that conversation.

The check I could not tune against

Twenty three-turn scenarios were authored by a different model, with graders frozen before replies existed. On the original run A0 and A5 both scored **17/20**, agreeing on 18/20; on the 13 unambiguous scenarios A0 scored 12/13 and A5 11/13. One A5 reply was then excluded by deterministic cleaning because a render failure was appended to a complete answer; excluding it leaves A5 at 16/19 overall and 10/12 on the scored scenarios. The result does not confirm the single-turn gain out of sample, and the set is too small to call it a regression.

Limits and failures

- The single-turn comparison is in-sample: A5 and its graders were iterated against the same scenarios. Twenty grader corrections were made, all after transcript inspection; correction 21 fixed a positive-control bug allowing correct three-term totals such as $99 + $150 + $50 = $299.
- Six multi-turn failures were authoring gaps: the brief omitted savings-club benefits. The remaining failures are agent misses on hours, overnight constraints, and some arithmetic/change-of-mind turns.
- One build showed website-accurate savings-club benefits that were absent from the brief while another build declined. The cause is unknown; the repository reports it as an open platform question, not as a proven defect.
- A human calibration reached κ=0.911 only after the rule was fitted to the same 30 transcripts; it is not fresh-sample agreement.

The complete evidence is in `RESULTS.md`, `RIGOR.md`, `FAILURES.md` and `FINDINGS-FOR-AGENTPLACE.md`. The live agent link is supplied separately.